

CSA15 – CLOUD COMPUTING AND BIG DATA ANALYTICS

Assignment Answers

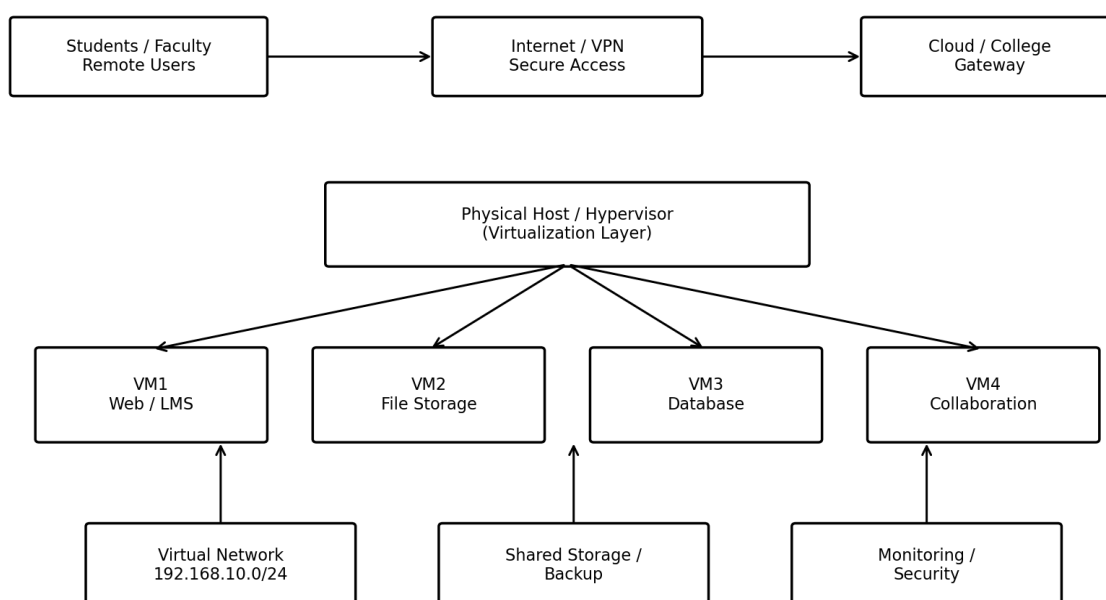
Design and Deployment of a Cloud-Based Virtualized Collaboration Platform for an Engineering College

Course Outcomes: CO1 – Cloud service models; CO2 – Virtualization and virtualized data center; CO3 – Cloud access and collaboration.

SDG Mapping: SDG 4 – Quality Education; SDG 8 – Decent Work and Economic Growth; SDG 9 – Industry, Innovation and Infrastructure.

Important note: This answer document follows the supplied assignment statement and assessment rubric point-to-point. The architecture and configuration below are a proposed implementation design. Screenshots/evidence that require an actually deployed environment are identified as evidence to capture rather than fabricated.

Virtualized Cloud Data-Center Architecture



1. Problem Statement – Requirement Analysis

The institution wants to migrate separate physical systems for application hosting, file storage and user access to a cloud-based environment supporting online learning, file sharing, application hosting and collaborative academic activities. The supplied problem identifies high infrastructure cost, inefficient resource utilization and difficulty handling periods of high demand as the main issues.

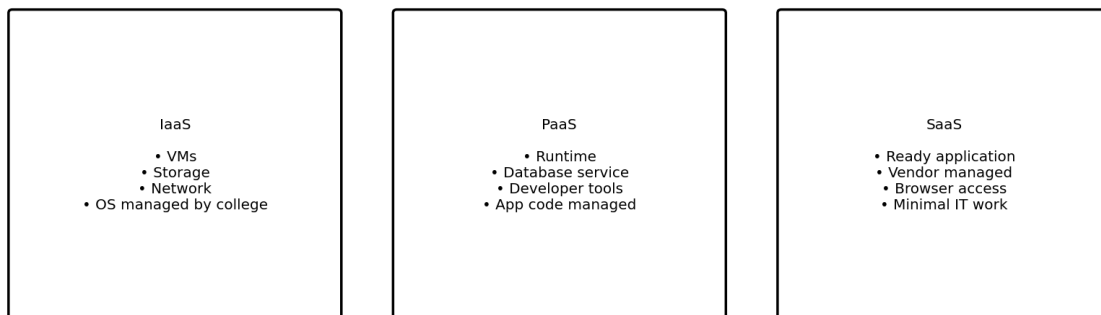
Requirement	Proposed cloud response	Reason
Online learning	LMS/Web application on VM or managed PaaS	Central access and easier scaling
File sharing/storage	Cloud file service or dedicated storage VM	Shared access, backup and controlled permissions
Application hosting	IaaS VM or PaaS runtime	Flexible hosting and deployment
User access	HTTPS portal + VPN/SSO/MFA	Secure remote access
Collaboration	SaaS collaboration suite / collaboration VM	Document sharing, editing and communication
High-demand periods	Elastic resources / additional VM capacity	Scalability without permanent overprovisioning
Resource management	Virtualization + monitoring	Higher utilization and centralized administration

The rubric specifically expects clear identification of institutional requirements and justified mapping to IaaS, PaaS and SaaS.

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2. Cloud Service Model Analysis (CO1)

Cloud Service Models - Responsibility View



Control decreases → IaaS → PaaS → SaaS ← Convenience increases

IaaS – Infrastructure as a Service

IaaS provides virtualized computing infrastructure such as virtual machines, storage and networking. For this college, IaaS is suitable when administrators need control over the operating system, VM resources, networking and installed software.

- **Benefits:** strong control, flexible VM sizing, suitable for legacy applications, easy creation of multiple isolated servers.
- **Limitations:** the institution still manages operating systems, patches, VM security and much of the network configuration.
- **Cost:** pay for allocated/consumed compute, storage and network resources; virtualization reduces the need to buy separate physical servers.

- Scalability: CPU, RAM, storage and additional VMs can be increased as demand grows.
- Accessibility: resources can be reached remotely through secured network access.
- Security: requires firewall rules, patching, least privilege, secure keys/passwords and monitoring.

PaaS – Platform as a Service

PaaS supplies a managed application platform/runtime. Developers focus mainly on application code while the platform provider manages much of the infrastructure and runtime.

- Benefits: faster application deployment, reduced OS administration, easier scaling and standardized development environment.
- Limitations: less infrastructure control and possible platform/vendor lock-in.
- Cost: reduces administration effort but managed platform usage may cost more per compute unit than raw infrastructure.
- Scalability: commonly supports easier application scaling than manually managing every VM.
- Accessibility: developers and authorized staff can work through web/API interfaces.
- Security: identity, application secrets, network restrictions and application-level security remain important.

SaaS – Software as a Service

SaaS delivers a ready-to-use application through a browser or client. For academic collaboration, SaaS can provide document sharing, collaborative editing, communication and calendars without the college maintaining the complete application stack.

- Benefits: minimal infrastructure management, rapid adoption, browser-based access and built-in collaboration.
- Limitations: least control over infrastructure and application internals; subscription and data-location/vendor dependency must be considered.
- Cost: generally subscription or usage based, avoiding large local deployment and maintenance costs.
- Scalability: provider handles most infrastructure scaling.
- Accessibility: strong for remote students and faculty when internet access is available.
- Security: use MFA, role-based access, sharing controls, audit logs and data protection policies.

The assignment rubric asks for analysis of characteristics, benefits, limitations, cost, scalability, accessibility and security.

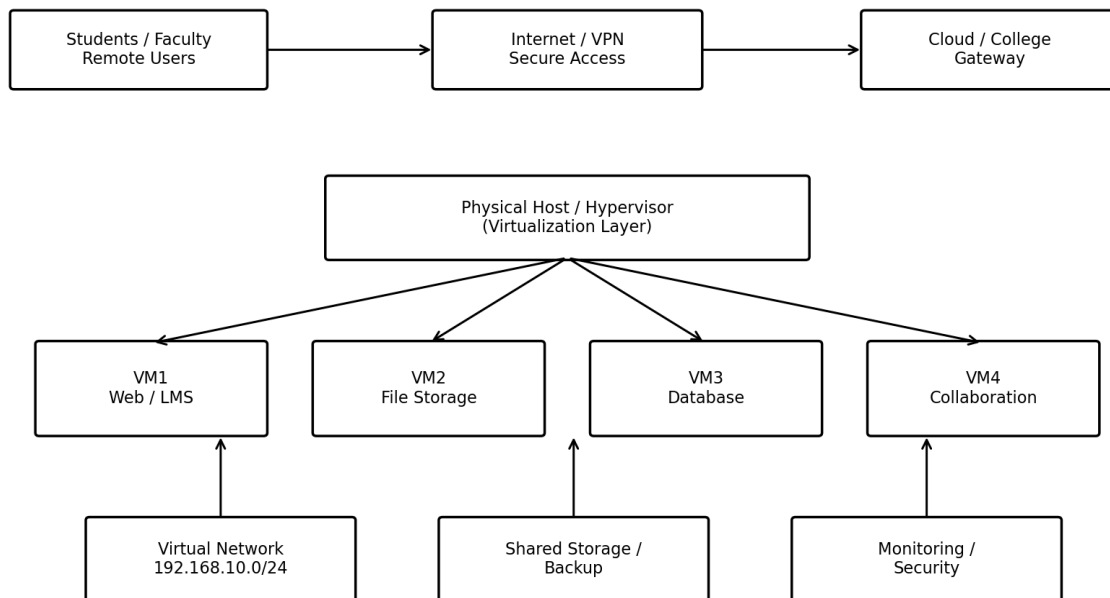
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3. Recommended IaaS / PaaS / SaaS Mapping

Institutional need	Model	Example role in solution	Justification
Virtual servers for LMS/web	IaaS	Web/LMS VM	Maximum OS and server control
Custom academic application	PaaS	Managed runtime	Faster development and deployment
Database-backed application	PaaS	Managed database	Less database administration
Document sharing	SaaS	Cloud office/storage suite	Ready-to-use collaboration
Email/communication	SaaS	Hosted mail/chat/meetings	No mail-server maintenance
Backup/storage	IaaS/SaaS	Object storage or managed drive	Durable centralized storage
Remote administration	IaaS + VPN	Admin access to VMs	Secure infrastructure management

4. Virtualized Data Center Architecture (CO2)

Virtualized Cloud Data-Center Architecture



The proposed small data center uses one physical host running a hypervisor. Multiple VMs represent logically separate infrastructure services. A virtual switch connects the VMs, while a firewall/VPN layer controls external access.

The rubric requires a clear architecture with well-defined VMs, services, resources and network connectivity.

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Suggested VM plan

VM	Service	vCPU	RAM	Storage	Example IP
VM1	Web/LMS	2	4 GB	40 GB	192.168.10.11
VM2	File storage	2	4 GB	100 GB	192.168.10.12
VM3	Database	2	4 GB	60 GB	192.168.10.13
VM4	Collaboration	2	4 GB	60 GB	192.168.10.14
Host	Hypervisor	Physical resources	According to hardware	Shared/local storage	192.168.10.10

Suggested virtualization tools: VirtualBox, VMware Workstation/ESXi, Hyper-V, or another suitable hypervisor. For a classroom demonstration, VirtualBox is simple; for a dedicated server, a type-1 hypervisor such as ESXi/Hyper-V can be considered.

Resource planning: Do not allocate all host CPU/RAM to VMs. Reserve capacity for the hypervisor and peak demand. Start with the above values and resize after monitoring actual utilization.

5. Virtual Machine Deployment & Resource Configuration (CO2)

Step 1 – Prepare the host

- Install and verify hardware virtualization support (Intel VT-x/AMD-V).
- Install the selected hypervisor.
- Create an internal/bridged/NAT network according to the lab environment.
- Create a dedicated VM storage location and backup location.

Step 2 – Create the VMs

- Create four VMs with the names VM1-Web, VM2-File, VM3-DB and VM4-Collab.
- Attach the selected Linux/Windows server ISO as appropriate.
- Assign vCPU, RAM, disk and virtual NIC according to the VM table.
- Install the operating system and update packages.
- Assign static IP addresses or DHCP reservations.

Step 3 – Install services

- VM1: install the LMS/web server and configure HTTPS.
- VM2: create controlled shared directories and storage quotas.
- VM3: install the required database engine and restrict database access to application VMs.
- VM4: deploy a collaboration application or connect the architecture to a SaaS collaboration service.

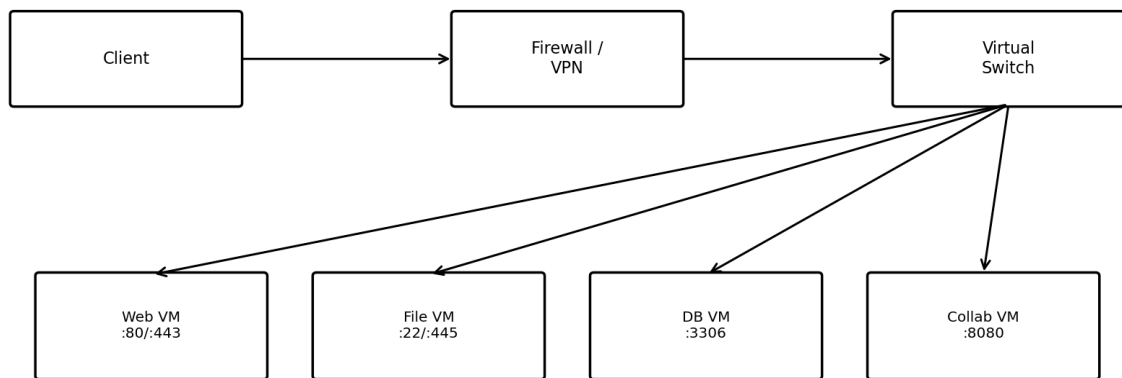
Step 4 – Verify resource configuration

- Check CPU and memory allocation in the hypervisor console.
- Check disk size and free space inside every VM.
- Check network adapter attachment and IP configuration.
- Record VM uptime, resource utilization and service status.

The rubric expects multiple VMs with appropriate CPU, memory, storage and networking resources and clear implementation evidence. ■filecite■turn0file0■L148-L183■

6. Network Configuration & Service Communication

VM Network and Service Communication



Example subnet: 192.168.10.0/24 | SSH only from admin network | HTTPS for users

Use a private virtual subnet such as 192.168.10.0/24. External users should reach public services through HTTPS. Administrative SSH/RDP should be restricted to the administrator network or VPN.

Test	Expected result
Ping between allowed VMs	Successful where ICMP is permitted
Web access to VM1	HTTPS page/LMS loads
File service from authorized client	Shared folder is accessible
VM3 database from VM1	Database connection succeeds
Unauthorized database access	Blocked by firewall
Admin SSH/RDP from VPN	Successful for authorized administrator only

The rubric specifically requires correct VM networking and demonstrated service-to-service communication.

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7. Cloud Access & Remote Resource Management (CO3)

A secure access mechanism should be used rather than exposing management ports directly to the public internet. The preferred approach is VPN-based administration combined with HTTPS for user-facing applications.

- User access: HTTPS to LMS/collaboration services.
- Administrator access: VPN → firewall → management network → SSH/RDP.
- Authentication: strong passwords plus MFA where supported.
- Authorization: least-privilege role assignments.
- Encryption: TLS for web traffic and encrypted VPN tunnel for remote administration.
- Monitoring: record login attempts, administrative actions and important service events.

The rubric requires secure remote access and an explanation of the selected mechanism.

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8. User Roles, Permissions & Security (CO3)

Role	Typical permissions	Restrictions
Student	Read course material; upload assigned work; collaborate	No server administration; limited sharing
Faculty	Create/update course files; collaborate; grade/feedback	No hypervisor administration
IT Administrator	Manage VMs, network, backups and monitoring	Use MFA/VPN; privileged account only when needed
Guest	View explicitly shared resources	No private files or management access

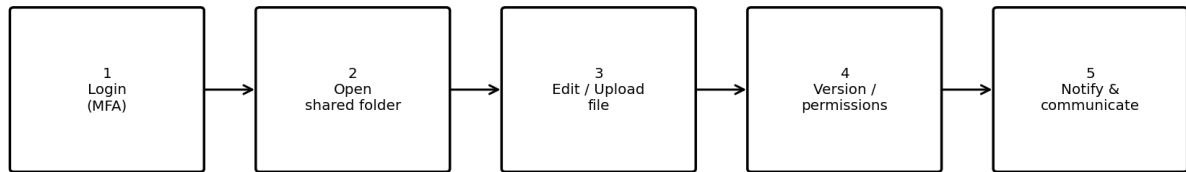
- Apply least privilege and role-based access control.
- Use separate administrator accounts rather than sharing one admin password.
- Disable unused services and close unnecessary firewall ports.
- Patch operating systems and applications regularly.
- Maintain backups and periodically test restoration.
- Use file permissions and sharing policies to prevent accidental data exposure.
- Keep audit logs and review suspicious login/activity patterns.

The rubric expects clearly defined roles, appropriate permissions, security and resource protection.

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9. Cloud Collaboration Services

Cloud Collaboration Workflow



Examples: shared documents • file access • collaborative editing • announcements • chat/video meetings

- Document sharing: centralized folders with role-based permissions.
- File access: browser/client access from authorized devices.
- Collaborative editing: multiple users can edit shared documents where supported.
- Communication: institutional chat, announcements and video meetings.
- Version control: retain previous versions to recover from accidental edits.
- Notifications: inform users of comments, assignments and document changes.

These collaboration functions directly address the supplied requirement for document sharing, file access, communication and collaborative editing. ■filecite■turn0file0■L37-L41■

10. Implementation Evidence / Screenshots to Capture

The assignment deliverables require screenshots/evidence of VM deployment and configuration.

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Evidence #	Screenshot / evidence	What it should prove
1	Hypervisor main console	Host and created VMs are visible
2	VM1 settings	CPU, RAM, disk and NIC
3	VM2 settings	CPU, RAM, disk and NIC
4	VM3 settings	CPU, RAM, disk and NIC
5	VM4 settings	CPU, RAM, disk and NIC
6	VM IP configuration	Static IP/subnet settings
7	VM-to-VM ping	Network communication
8	Web/LMS page	Application hosting works
9	Shared folder/file access	File service works
10	Database connection	Application-to-database communication
11	VPN/admin login	Secure remote administration
12	User/role permissions	Access control works

Do not submit fabricated screenshots. Capture the real console/terminal/browser output after completing the deployment.

11. Comparison with Traditional Infrastructure

Factor	Traditional physical setup	Proposed virtualized cloud setup
Resource utilization	Often low when separate servers are idle	VM resources can be consolidated and adjusted
Scalability	Requires purchasing/installing hardware	VMs/resources can be resized or added faster
Accessibility	Often dependent on campus network	Secure internet/VPN access supports remote users
Cost	Higher hardware, power and maintenance burden	Less physical hardware; usage/hosting costs remain
Security	Physical isolation can be strong but patching is decentralized	Centralized controls help, but misconfiguration can affect many services
Collaboration	May require separate systems	Cloud tools support shared files and collaborative editing
Management	Many physical systems	Centralized VM and monitoring management
Availability	Hardware failure can stop a service	VM backup/recovery and redundancy can improve resilience

12. Advantages and Limitations of the Proposed Environment

Advantages

- Better resource utilization through server consolidation.
- Scalability by changing VM resources or adding VMs.
- Centralized management of compute, storage and networking.
- Remote accessibility through secure cloud/VPN mechanisms.
- Supports collaboration and shared academic resources.
- Easier backup, cloning and recovery of VMs than rebuilding separate physical servers.
- Clear separation of services improves administration and troubleshooting.

Limitations / Challenges

- Initial virtualization/cloud design requires technical knowledge.
- Poorly secured VMs or exposed management interfaces can create security risks.
- Cloud services depend on reliable network connectivity.
- Managed SaaS/PaaS can create vendor dependency.
- Resource over-allocation can still cause performance problems.
- Licensing, subscription and data-governance costs must be considered.
- A single physical host remains a potential single point of failure unless redundancy is introduced.

13. Cost, Scalability, Accessibility and Security Analysis

Cost: The proposed design reduces dependence on multiple physical servers and can lower power, space and maintenance requirements. However, cloud compute, storage, bandwidth, backup and SaaS/PaaS subscriptions still generate recurring costs.

Scalability: Vertical scaling increases VM CPU/RAM; horizontal scaling adds instances. During examination or admission peaks, extra application capacity can be provisioned and later released.

Accessibility: Browser-based services can support students and faculty outside campus. Secure authentication, HTTPS and VPN are essential.

Security: Security should be layered: identity and MFA, least privilege, firewall rules, network segmentation, patching, encryption, backups, logging and monitoring.

The rubric explicitly calls for reflection on scalability, resource utilization, accessibility, cost, security, collaboration and challenges. ■filecite■turn0file0■L305-L324■

14. Learning Outcomes / Reflection

- CO1: The solution distinguishes IaaS, PaaS and SaaS and maps each to an institutional requirement.
- CO2: The design demonstrates how a hypervisor can host multiple service-specific VMs and how their CPU, RAM, storage and networking are planned.
- CO3: The solution demonstrates secure cloud access, role-based permissions and collaboration workflows.
- The exercise shows why virtualization improves utilization and flexibility compared with maintaining isolated physical servers.
- The exercise also highlights that cloud adoption does not remove security or management responsibilities; it changes where and how those responsibilities are performed.

15. Conclusion

The proposed cloud-based virtualized collaboration platform addresses the college's need for online learning, file sharing, application hosting, remote access and academic collaboration. IaaS is appropriate where infrastructure control is required, PaaS is appropriate for managed application development/deployment, and SaaS is appropriate for ready-to-use collaboration and communication. A small hypervisor-based data center with separate VMs provides service isolation, centralized management and flexible resource allocation. Secure VPN/HTTPS access, role-based permissions and layered security controls protect resources while enabling remote academic work.

16. Deliverables Checklist

- ✓ Architecture Design – included in this answer document.
- ✓ Virtualized Data Center Implementation & Results – deployment procedure and expected verification included; add actual results/screenshots after implementation.
- ✓ Cloud Service Model Analysis – IaaS/PaaS/SaaS analysis included.
- ✓ Screenshots/Evidence – capture the listed real screenshots during implementation.
- ✓ Comparison and Analysis Report – included.
- ✓ GitHub Upload – upload the project files, configuration notes, architecture diagram and evidence folder to the required repository.

This structure is aligned to the supplied assignment deliverables and the 100-mark assessment rubric.

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